DATABASE RELATIONSHIP ANALYSIS DOCUMENT

NAILVERSE

Not just NAILS.. IT's the WHOLE MOOD

Online Nail Art & Nail Beauty Online Platform

WEEK - 8

---

8. Database Relationship Analysis using Joins

Scenario

The details of Nailverse are spread over several tables: customers, appointments, the services inside each appointment, and payments. On their own these tables only answer small questions. The management team wants to connect them so that one result can show who booked what, when it was booked, how much it cost and whether it has been paid.

This is done with SQL joins. Each join type is tried on the Nailverse tables to see how matching and non-matching rows behave. The data used here includes Divya Menon, a registered customer with no appointment, and the Gel Extension service, which nobody has booked yet.

1. Implement INNER JOIN

Concept:

INNER JOIN returns only the rows that have a match in both tables.

Tasks:

1. Display customer details along with their appointments.

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         a.Appointment_ID,
  3         a.Total_Amount
  4  FROM Customer c
  5  INNER JOIN Appointment a
  6      ON c.Customer_ID = a.Customer_ID
  7  ORDER BY a.Appointment_ID;

CUSTOMER_NAME                          APPOINT TOTAL_AMOUNT
-------------------------------------- ------- ------------
MEERA IYER                             A001             499
KAVYA RAO                              A002             650
ANANYA  SURESH                         A003             459
PRIYA NAIR                             A004             999
NISHA KUMARI                           A005             799
VARSHA RAJAN                           A006            1299
MOSHAMI JUN                            A007             499
HARI KUMAR                             A008             650
SRI VALLI                              A009             459
NILLA VEERA                            A010             999

10 rows selected.
15 rows selected.
```

2. Display appointment details with payment information.

```
SQL> SELECT a.Appointment_ID,
  2         a.Total_Amount,
  3         a.Appointment_Date,
  4         a.Customer_ID,
  5         p.Payment_Date,
  6         p.Payment_Mode,
  7         p.Payment_Status
  8  FROM Payment p
  9  JOIN Appointment a
 10      ON p.Appointment_ID = a.Appointment_ID
 11  ORDER BY p.Payment_ID;

APPOINT TOTAL_AMOUNT APPOINTME CUSTOMER_ID          PAYMENT_D PAYMENT_MODE
------- ------------ --------- -------------------- --------- ---------------
PAYMENT_ST
----------
A001             499 24-OCT-26 CU-1001              01-SEP-26 Credit Card
COMPLETED

A002             650 25-JAN-26 CU-1002              25-JAN-26 UPI
COMPLETED

A003             459 26-OCT-26 CU-1003              26-OCT-26 Debit Card
FAILED


APPOINT TOTAL_AMOUNT APPOINTME CUSTOMER_ID          PAYMENT_D PAYMENT_MODE
------- ------------ --------- -------------------- --------- ---------------
PAYMENT_ST
----------
A004             999 27-DEC-26 CU-1004              27-DEC-26 Net Banking
COMPLETED

A005             799 28-JUN-26 CU-1005              28-JUN-26 UPI
COMPLETED

A006            1299 29-OCT-26 CU-1006              29-OCT-26 Debit Card
FAILED


APPOINT TOTAL_AMOUNT APPOINTME CUSTOMER_ID          PAYMENT_D PAYMENT_MODE
------- ------------ --------- -------------------- --------- ---------------
PAYMENT_ST
----------
A007             499 30-MAY-26 CU-1007              30-MAY-26 UPI
COMPLETED

A008             650 31-OCT-26 CU-1008              31-OCT-26 Credit Card
COMPLETED

A009             459 01-FEB-26 CU-1009              01-FEB-26 Cash
FAILED


APPOINT TOTAL_AMOUNT APPOINTME CUSTOMER_ID          PAYMENT_D PAYMENT_MODE
------- ------------ --------- -------------------- --------- ---------------
PAYMENT_ST
----------
A010             999 02-APR-26 CU-1010              02-APR-26 UPI
COMPLETED


10 rows selected.
```

Appointment A006 appears twice because it has two payment records: a failed attempt and the successful retry.

3. Retrieve the services booked by customers.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         a.Appointment_ID,
  4         s.Service_Name,
  5         ad.Quantity,
  6         ad.Price,
  7         ad.Quantity * ad.Price AS Total_Amount
  8  FROM Customer c
  9  JOIN Appointment a
 10      ON c.Customer_ID = a.Customer_ID
 11  JOIN Appointment_Details ad
 12      ON a.Appointment_ID = ad.Appointment_ID
 13  JOIN Service s
 14      ON ad.Service_ID = s.Service_ID
 15  ORDER BY c.Customer_ID, a.Appointment_Date;

CUSTOMER_ID          CUSTOMER_NAME                             APPOINT
-------------------- ----------------------------------------- -------
SERVICE_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------
CU-1001              MEERA IYER                                A001
CLASSIC MAINCURE                                            1        499
         499

CU-1002              KAVYA RAO                                 A002
CLASSIC PEDICURE                                            1        650
         650

CUSTOMER_ID          CUSTOMER_NAME                             APPOINT
-------------------- ----------------------------------------- -------
SERVICE_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------

CU-1003              ANANYA  SURESH                            A003
NAIL EXTENSIONS                                             1        459
         459

CU-1004              PRIYA NAIR                                A004
CHROME NAIL ART                                             1        999

CUSTOMER_ID          CUSTOMER_NAME                             APPOINT
-------------------- ----------------------------------------- -------
SERVICE_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------
         999

CU-1005              NISHA KUMARI                              A005
SPA PEDICURE                                                1        799
         799
```

```
CU-1006             VARSHA RAJAN                            A006

CUSTOMER_ID         CUSTOMER_NAME                           APPOINT
------------------- --------------------------------------- -------
SERVICE_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------
Chrome Nail Art                                             1       1299
        1299

CU-1007             MOSHAMI JUN                             A007
CLASSIC MAINCURE                                            1        499
         499


CUSTOMER_ID         CUSTOMER_NAME                           APPOINT
------------------- --------------------------------------- -------
SERVICE_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------
CU-1008             HARI KUMAR                              A008
CLASSIC PEDICURE                                            1        650
         650

CU-1009             SRI VALLI                               A009
NAIL EXTENSIONS                                             1        459
         459

CUSTOMER_ID         CUSTOMER_NAME                           APPOINT
------------------- --------------------------------------- -------
SERVICE_NAME                                       QUANTITY      PRICE
-------------------------------------------------- ---------- ----------
TOTAL_AMOUNT
------------

CU-1010             NILLA VEERA                             A010
CHROME NAIL ART                                             1        999
         999


10 rows selected.
```

2. Implement LEFT JOIN

Concept:

LEFT JOIN returns every row of the left table and the matching rows of the right table. Where there is no match, the right-side columns are empty (NULL).

Tasks:

1. Display all customers, including customers who have not booked an appointment.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         a.Appointment_ID,
  4         a.Appointment_Date,
  5         a.Total_Amount,
  6         a.Appointment_Status
  7  FROM Customer c
  8  LEFT JOIN Appointment a
  9      ON c.Customer_ID = a.Customer_ID
 10  ORDER BY c.Customer_ID, a.Appointment_Date;

CUSTOMER_ID          CUSTOMER_NAME                             APPOINT APPOINTME
-------------------- ----------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1001              MEERA IYER                                A001    24-OCT-26
         499 COMPLETED

CU-1002              KAVYA RAO                                 A002    25-JAN-26
         650 PENDING

CU-1003              ANANYA  SURESH                            A003    26-OCT-26
         459 PENDING


CUSTOMER_ID          CUSTOMER_NAME                             APPOINT APPOINTME
-------------------- ----------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1004              PRIYA NAIR                                A004    27-DEC-26
         999 COMPLETED

CU-1005              NISHA KUMARI                              A005    28-JUN-26
         799 PENDING

CU-1006              VARSHA RAJAN                              A006    29-OCT-26
        1299 PENDING


CUSTOMER_ID          CUSTOMER_NAME                             APPOINT APPOINTME
-------------------- ----------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1007              MOSHAMI JUN                               A007    30-MAY-26
         499 COMPLETED

CU-1008              HARI KUMAR                                A008    31-OCT-26
         650 PENDING

CU-1009              SRI VALLI                                 A009    01-FEB-26
         459 PENDING


CUSTOMER_ID          CUSTOMER_NAME                             APPOINT APPOINTME
-------------------- ----------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1010              NILLA VEERA                               A010    02-APR-26
         999 COMPLETED


10 rows selected.
```

2. Find the customers who have never booked an appointment.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         c.Email,
  4         c.Phone
  5  FROM Customer c
  6  LEFT JOIN Appointment a
  7      ON c.Customer_ID = a.Customer_ID
  8  WHERE a.Appointment_ID IS NULL
  9  ORDER BY c.Customer_ID;

no rows selected
```

3. Display all services, including services that have never been booked.

```
SQL> SELECT s.Service_ID,
  2         s.Service_Name,
  3         s.Price
  4  FROM Service s
  5  LEFT JOIN Appointment_Details ad
  6       ON s.Service_ID = ad.Service_ID
  7  WHERE ad.Service_ID IS NULL
  8  ORDER BY s.Service_ID;

no rows selected
```

The query keeps only the services whose booking columns are empty, that is, the services with no bookings.

3. Implement RIGHT JOIN

Concept:

RIGHT JOIN returns every row of the right table and the matching rows of the left table.

Tasks:

1. Display all appointments with customer information.

```
SQL> SELECT a.Appointment_ID,
  2         a.Appointment_Date,
  3         a.Total_Amount,
  4         a.Appointment_Status,
  5         c.Customer_ID,
  6         c.First_Name || ' ' || c.Last_Name AS Customer_Name
  7  FROM Customer c
  8  RIGHT JOIN Appointment a
  9      ON c.Customer_ID = a.Customer_ID
 10  ORDER BY a.Appointment_Date;

APPOINT APPOINTME TOTAL_AMOUNT APPOINTMENT_STATUS   CUSTOMER_ID
------- --------- ------------ -------------------- --------------------
CUSTOMER_NAME
----------------------------------------
A002    25-JAN-26          650 PENDING              CU-1002
KAVYA RAO

A009    01-FEB-26          459 PENDING              CU-1009
SRI VALLI

A010    02-APR-26          999 COMPLETED            CU-1010
NILLA VEERA


APPOINT APPOINTME TOTAL_AMOUNT APPOINTMENT_STATUS   CUSTOMER_ID
------- --------- ------------ -------------------- --------------------
CUSTOMER_NAME
----------------------------------------
A007    30-MAY-26          499 COMPLETED            CU-1007
MOSHAMI JUN

A005    28-JUN-26          799 PENDING              CU-1005
NISHA KUMARI

A001    24-OCT-26          499 COMPLETED            CU-1001
MEERA IYER


APPOINT APPOINTME TOTAL_AMOUNT APPOINTMENT_STATUS   CUSTOMER_ID
------- --------- ------------ -------------------- --------------------
CUSTOMER_NAME
----------------------------------------
A003    26-OCT-26          459 PENDING              CU-1003
ANANYA  SURESH

A006    29-OCT-26         1299 PENDING              CU-1006
VARSHA RAJAN

A008    31-OCT-26          650 PENDING              CU-1008
HARI KUMAR


APPOINT APPOINTME TOTAL_AMOUNT APPOINTMENT_STATUS   CUSTOMER_ID
------- --------- ------------ -------------------- --------------------
CUSTOMER_NAME
----------------------------------------
A004    27-DEC-26          999 COMPLETED            CU-1004
PRIYA NAIR


10 rows selected.
```

2. Find the appointments where customer details are missing.

```
10 rows selected.

SQL> SELECT a.Appointment_ID,
  2         a.Appointment_Date,
  3         a.Total_Amount,
  4         a.Appointment_Status
  5  FROM Customer c
  6  RIGHT JOIN Appointment a
  7      ON c.Customer_ID = a.Customer_ID
  8  WHERE c.Customer_ID IS NULL;

no rows selected
```

No rows are returned. The foreign key from Appointment to Customer does not allow an appointment without a valid customer.

3. Display all payment records with appointment details.

```
SQL> SELECT p.Payment_ID,
  2         p.Payment_Date,
  3         p.Payment_Mode,
  4         p.Payment_Status,
  5         p.Transaction_Amount AS Amount,
  6         a.Appointment_ID,
  7         a.Appointment_Date,
  8         a.Total_Amount
  9  FROM Appointment a
 10  RIGHT JOIN Payment p
 11      ON a.Appointment_ID = p.Appointment_ID
 12  ORDER BY p.Payment_Date, p.Payment_ID;

PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST     AMOUNT APPOINT APPOINTME
------- --------- --------------- ---------- ---------- ------- ---------
TOTAL_AMOUNT
------------
PAY0002 25-JAN-26 UPI             COMPLETED        1200 A002    25-JAN-26
         650

PAY0009 01-FEB-26 Cash            FAILED           1100 A009    01-FEB-26
         459

PAY0010 02-APR-26 UPI             COMPLETED        2500 A010    02-APR-26
         999


PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST     AMOUNT APPOINT APPOINTME
------- --------- --------------- ---------- ---------- ------- ---------
TOTAL_AMOUNT
------------
PAY0007 30-MAY-26 UPI             COMPLETED        1800 A007    30-MAY-26
         499

PAY0005 28-JUN-26 UPI             COMPLETED        2000 A005    28-JUN-26
         799

PAY0001 01-SEP-26 Credit Card     COMPLETED         999 A001    24-OCT-26
         499


PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST     AMOUNT APPOINT APPOINTME
------- --------- --------------- ---------- ---------- ------- ---------
TOTAL_AMOUNT
------------
PAY0003 26-OCT-26 Debit Card      FAILED           1500 A003    26-OCT-26
         459

PAY0006 29-OCT-26 Debit Card      FAILED            750 A006    29-OCT-26
        1299

PAY0008 31-OCT-26 Credit Card     COMPLETED         950 A008    31-OCT-26
         650


PAYMENT PAYMENT_D PAYMENT_MODE    PAYMENT_ST     AMOUNT APPOINT APPOINTME
------- --------- --------------- ---------- ---------- ------- ---------
TOTAL_AMOUNT
------------
PAY0004 27-DEC-26 Net Banking     COMPLETED         800 A004    27-DEC-26
         999


10 rows selected.
```

4. Implement FULL OUTER JOIN

Concept:

FULL OUTER JOIN returns all rows from both tables, matched wherever possible and with empty columns where there is no match.

1. Display all customers and all appointments, including those without a match on either side.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         a.Appointment_ID,
  4         a.Appointment_Date
  5  FROM Customer c
  6  FULL OUTER JOIN Appointment a
  7      ON c.Customer_ID = a.Customer_ID
  8  ORDER BY c.Customer_ID, a.Appointment_ID;

CUSTOMER_ID          CUSTOMER_NAME                          APPOINT APPOINTME
-------------------- -------------------------------------- ------- ---------
CU-1001              MEERA IYER                             A001    24-OCT-26
CU-1002              KAVYA RAO                              A002    25-JAN-26
CU-1003              ANANYA  SURESH                         A003    26-OCT-26
CU-1004              PRIYA NAIR                             A004    27-DEC-26
CU-1005              NISHA KUMARI                           A005    28-JUN-26
CU-1006              VARSHA RAJAN                           A006    29-OCT-26
CU-1007              MOSHAMI JUN                            A007    30-MAY-26
CU-1008              HARI KUMAR                             A008    31-OCT-26
CU-1009              SRI VALLI                              A009    01-FEB-26
CU-1010              NILLA VEERA                            A010    02-APR-26

10 rows selected.
```

5. Retrieve Complete Appointment Details

Task: Generate a complete appointment report by combining:

- Customer name
- Service name
- Quantity
- Appointment date
- Total amount
- Payment status

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         s.Service_Name,
  3         ad.Quantity,
  4         a.Appointment_Date,
  5         a.Total_Amount,
  6         pm.Payment_Status
  7  FROM Customer c
  8  JOIN Appointment a
  9      ON c.Customer_ID = a.Customer_ID
 10  JOIN Appointment_Details ad
 11      ON a.Appointment_ID = ad.Appointment_ID
 12  JOIN Service s
 13      ON ad.Service_ID = s.Service_ID
 14  LEFT JOIN Payment pm
 15      ON a.Appointment_ID = pm.Appointment_ID
 16  ORDER BY a.Appointment_Date, c.Customer_ID, pm.Payment_ID;

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                    QUANTITY APPOINTME
---------------------------------------------- ---------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
KAVYA RAO
CLASSIC PEDICURE                                         1 25-JAN-26
         650 COMPLETED

SRI VALLI
NAIL EXTENSIONS                                          1 01-FEB-26
         459 FAILED

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                    QUANTITY APPOINTME
---------------------------------------------- ---------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------

NILLA VEERA
CHROME NAIL ART                                          1 02-APR-26
         999 COMPLETED

MOSHAMI JUN
CLASSIC MAINCURE                                         1 30-MAY-26
```

```
CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                     QUANTITY APPOINTME
------------------------------------------------ ---------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
         499 COMPLETED

NISHA KUMARI
SPA PEDICURE                                              1 28-JUN-26
         799 COMPLETED

MEERA IYER

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                     QUANTITY APPOINTME
------------------------------------------------ ---------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
CLASSIC MAINCURE                                          1 24-OCT-26
         499 COMPLETED

ANANYA  SURESH
NAIL EXTENSIONS                                           1 26-OCT-26
         459 FAILED


CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                     QUANTITY APPOINTME
------------------------------------------------ ---------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------
VARSHA RAJAN
Chrome Nail Art                                           1 29-OCT-26
        1299 FAILED

HARI KUMAR
CLASSIC PEDICURE                                          1 31-OCT-26
         650 COMPLETED

CUSTOMER_NAME
----------------------------------------
SERVICE_NAME                                     QUANTITY APPOINTME
------------------------------------------------ ---------- ---------
TOTAL_AMOUNT PAYMENT_ST
------------ ----------

PRIYA NAIR
CHROME NAIL ART                                           1 27-DEC-26
         999 COMPLETED


10 rows selected.
```

6. Display Customer Booking History

Tasks: Write queries to find:

1. All services booked by one customer (Meera Iyer, CU-1001).

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         a.Appointment_ID,
  3         a.Appointment_Date,
  4         s.Service_Name,
  5         ad.Quantity,
  6         ad.Price
  7  FROM Customer c
  8  JOIN Appointment a
  9      ON c.Customer_ID = a.Customer_ID
 10  JOIN Appointment_Details ad
 11      ON a.Appointment_ID = ad.Appointment_ID
 12  JOIN Service s
 13      ON ad.Service_ID = s.Service_ID
 14  WHERE c.Customer_ID = 'CU-1001'
 15  ORDER BY a.Appointment_Date;

CUSTOMER_NAME                             APPOINT APPOINTME
----------------------------------------- ------- ---------
SERVICE_NAME                                    QUANTITY      PRICE
------------------------------------------------- ---------- ----------
MEERA IYER                                A001    24-OCT-26
CLASSIC MAINCURE                                         1        499
```

2. Total amount spent by each customer.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         ROUND(SUM(a.Total_Amount), 2) AS Total_Amount_Spent
  4  FROM Customer c
  5  JOIN Appointment a
  6      ON c.Customer_ID = a.Customer_ID
  7  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  8  ORDER BY Total_Amount_Spent DESC;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_AMOUNT_SPENT
------------------
CU-1006              VARSHA RAJAN
              1299

CU-1004              PRIYA NAIR
               999

CU-1010              NILLA VEERA
               999


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_AMOUNT_SPENT
------------------
CU-1005              NISHA KUMARI
               799

CU-1002              KAVYA RAO
               650

CU-1008              HARI KUMAR
               650


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_AMOUNT_SPENT
------------------
CU-1007              MOSHAMI JUN
               499

CU-1001              MEERA IYER
               499

CU-1003              ANANYA  SURESH
               459


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
TOTAL_AMOUNT_SPENT
------------------
CU-1009              SRI VALLI
               459


10 rows selected.
```

3. Number of appointments booked by each customer.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         COUNT(a.Appointment_ID) AS Number_of_Appointments
  4  FROM Customer c
  5  LEFT JOIN Appointment a
  6      ON c.Customer_ID = a.Customer_ID
  7  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  8  ORDER BY Number_of_Appointments DESC, c.Customer_ID;

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1001              MEERA IYER
                     1

CU-1002              KAVYA RAO
                     1

CU-1003              ANANYA  SURESH
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1004              PRIYA NAIR
                     1

CU-1005              NISHA KUMARI
                     1

CU-1006              VARSHA RAJAN
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1007              MOSHAMI JUN
                     1

CU-1008              HARI KUMAR
                     1

CU-1009              SRI VALLI
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1010              NILLA VEERA
                     1


10 rows selected.
```

4. Latest booking details of each customer.

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         a.Appointment_ID,
  4         a.Appointment_Date AS Latest_Booking,
  5         a.Total_Amount,
  6         a.Appointment_Status
  7  FROM Customer c
  8  JOIN Appointment a
  9      ON c.Customer_ID = a.Customer_ID
 10  WHERE a.Appointment_Date = (
 11      SELECT MAX(a2.Appointment_Date)
 12      FROM Appointment a2
 13      WHERE a2.Customer_ID = c.Customer_ID
 14  )
 15  ORDER BY c.Customer_ID;

CUSTOMER_ID          CUSTOMER_NAME                          APPOINT LATEST_BO
-------------------- -------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1001              MEERA IYER                             A001    24-OCT-26
         499 COMPLETED

CU-1002              KAVYA RAO                              A002    25-JAN-26
         650 PENDING

CU-1003              ANANYA  SURESH                         A003    26-OCT-26
         459 PENDING


CUSTOMER_ID          CUSTOMER_NAME                          APPOINT LATEST_BO
-------------------- -------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1004              PRIYA NAIR                             A004    27-DEC-26
         999 COMPLETED

CU-1005              NISHA KUMARI                           A005    28-JUN-26
         799 PENDING

CU-1006              VARSHA RAJAN                           A006    29-OCT-26
        1299 PENDING


CUSTOMER_ID          CUSTOMER_NAME                          APPOINT LATEST_BO
-------------------- -------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1007              MOSHAMI JUN                            A007    30-MAY-26
         499 COMPLETED

CU-1008              HARI KUMAR                             A008    31-OCT-26
         650 PENDING

CU-1009              SRI VALLI                              A009    01-FEB-26
         459 PENDING


CUSTOMER_ID          CUSTOMER_NAME                          APPOINT LATEST_BO
-------------------- -------------------------------------- ------- ---------
TOTAL_AMOUNT APPOINTMENT_STATUS
------------ --------------------
CU-1010              NILLA VEERA                            A010    02-APR-26
         999 COMPLETED


10 rows selected.
```

7. Generate Multi-Table Business Reports

Report 1: Customer Appointment Report

Display:

- Customer name
- Appointment ID
- Appointment date
- Appointment status

```
SQL> SELECT c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  2         a.Appointment_ID,
  3         a.Appointment_Date,
  4         a.Appointment_Status
  5  FROM Customer c
  6  JOIN Appointment a
  7      ON c.Customer_ID = a.Customer_ID
  8  ORDER BY a.Appointment_Date;

CUSTOMER_NAME                          APPOINT APPOINTME APPOINTMENT_STATUS
-------------------------------------- ------- --------- --------------------
KAVYA RAO                              A002    25-JAN-26 PENDING
SRI VALLI                              A009    01-FEB-26 PENDING
NILLA VEERA                            A010    02-APR-26 COMPLETED
MOSHAMI JUN                            A007    30-MAY-26 COMPLETED
NISHA KUMARI                           A005    28-JUN-26 PENDING
MEERA IYER                             A001    24-OCT-26 COMPLETED
ANANYA  SURESH                         A003    26-OCT-26 PENDING
VARSHA RAJAN                           A006    29-OCT-26 PENDING
HARI KUMAR                             A008    31-OCT-26 PENDING
PRIYA NAIR                             A004    27-DEC-26 COMPLETED

10 rows selected.
```

Report 2: Sales Report

Display:

- Service name
- Quantity sold
- Total revenue

```
SQL> SELECT s.Service_Name,
  2         SUM(ad.Quantity) AS Quantity_Sold,
  3         SUM(ad.Quantity * ad.Price) AS Total_Revenue
  4  FROM Service s
  5  JOIN Appointment_Details ad
  6      ON s.Service_ID = ad.Service_ID
  7  GROUP BY s.Service_ID, s.Service_Name
  8  ORDER BY Total_Revenue DESC, s.Service_Name;

SERVICE_NAME                                QUANTITY_SOLD TOTAL_REVENUE
------------------------------------------- ------------- -------------
CHROME NAIL ART                                         2          1998
CLASSIC PEDICURE                                        2          1300
Chrome Nail Art                                         1          1299
CLASSIC MAINCURE                                        2           998
NAIL EXTENSIONS                                         2           918
SPA PEDICURE                                            1           799

6 rows selected.
```

Report 3: Payment Analysis Report

Display:

- Payment mode
- Number of transactions
- Successful payments

```
SQL> SELECT Payment_Mode,
  2         COUNT(*) AS Number_of_Transactions,
  3         SUM(
  4             CASE
  5                 WHEN Payment_Status = 'COMPLETED' THEN 1
  6                 ELSE 0
  7             END
  8         ) AS Successful_Payments
  9  FROM Payment
 10  GROUP BY Payment_Mode
 11  ORDER BY Number_of_Transactions DESC, Payment_Mode;

PAYMENT_MODE    NUMBER_OF_TRANSACTIONS SUCCESSFUL_PAYMENTS
--------------- ---------------------- ------------------
UPI                                  4                  4
Credit Card                          2                  2
Debit Card                           2                  0
Cash                                 1                  0
Net Banking                          1                  1
```

Report 4: Customer Booking Analysis

Find:

- Top booking customers (by total spending)

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         SUM(a.Total_Amount) AS Total_Spending
  4  FROM Customer c
  5  JOIN Appointment a
  6      ON c.Customer_ID = a.Customer_ID
  7  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  8  ORDER BY Total_Spending DESC;

CUSTOMER_ID          CUSTOMER_NAME                          TOTAL_SPENDING
-------------------- -------------------------------------- --------------
CU-1006              VARSHA RAJAN                                     1299
CU-1004              PRIYA NAIR                                        999
CU-1010              NILLA VEERA                                       999
CU-1005              NISHA KUMARI                                      799
CU-1002              KAVYA RAO                                         650
CU-1008              HARI KUMAR                                        650
CU-1007              MOSHAMI JUN                                       499
CU-1001              MEERA IYER                                        499
CU-1003              ANANYA  SURESH                                    459
CU-1009              SRI VALLI                                         459

10 rows selected.
```

- Customers with the maximum number of appointments

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         COUNT(a.Appointment_ID) AS Number_of_Appointments
  4  FROM Customer c
  5  JOIN Appointment a
  6      ON c.Customer_ID = a.Customer_ID
  7  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  8  HAVING COUNT(a.Appointment_ID) = (
  9      SELECT MAX(Appointment_Count)
 10      FROM (
 11          SELECT COUNT(*) AS Appointment_Count
 12          FROM Appointment
 13          GROUP BY Customer_ID
 14      )
 15  );

CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1004              PRIYA NAIR
                     1

CU-1006              VARSHA RAJAN
                     1

CU-1007              MOSHAMI JUN
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1009              SRI VALLI
                     1

CU-1002              KAVYA RAO
                     1

CU-1010              NILLA VEERA
                     1
```

-

```
CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1009              SRI VALLI
                     1

CU-1002              KAVYA RAO
                     1

CU-1010              NILLA VEERA
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1005              NISHA KUMARI
                     1

CU-1003              ANANYA  SURESH
                     1

CU-1008              HARI KUMAR
                     1


CUSTOMER_ID          CUSTOMER_NAME
-------------------- ----------------------------------------
NUMBER_OF_APPOINTMENTS
----------------------
CU-1001              MEERA IYER
                     1


10 rows selected.
```

- Customers with the highest spending

```
SQL> SELECT c.Customer_ID,
  2         c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  3         SUM(a.Total_Amount) AS Total_Spending
  4  FROM Customer c
  5  JOIN Appointment a
  6      ON c.Customer_ID = a.Customer_ID
  7  GROUP BY c.Customer_ID, c.First_Name, c.Last_Name
  8  HAVING SUM(a.Total_Amount) = (
  9      SELECT MAX(Total_Spending)
 10      FROM (
 11          SELECT SUM(Total_Amount) AS Total_Spending
 12          FROM Appointment
 13          GROUP BY Customer_ID
 14      )
 15  );

CUSTOMER_ID          CUSTOMER_NAME                             TOTAL_SPENDING
-------------------- ---------------------------------------- --------------
CU-1006              VARSHA RAJAN                                        1299
```